

PULLMAN-MOSCOW, WA, USA

WMO: 727857

Lat: 46.744N

Lon: 117.109W

Elev: 778

StdP: 92.33

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94129

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.8	-10.0	-17.7	0.9	-12.5	-13.8	1.2	-8.7	13.9	3.8	11.9	3.1	1.8	90	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	33.6	17.2	32.0	16.9	29.8	16.2	19.4	29.4	18.0	28.4	17.0	27.5	2.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.4	23.7	14.0	11.0	22.0	12.8	10.1	20.7	58.4	29.3	53.8	28.1	50.4	27.7	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.1	8.2	-19.5	36.8	4.7	1.7	-22.9	38.1	-25.6	39.1	-28.3	40.0	-31.7	41.2		
(5)			-20.0	20.8	4.5	2.2	-23.3	22.4	-25.9	23.7	-28.5	24.9	-31.7	26.5		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.2	1.3	4.8	8.0	12.1	15.2	19.6	19.4	15.2	8.9	3.5	-0.5
(7)		DBStd	8.04	4.76	4.51	3.70	3.56	3.95	3.60	3.27	3.47	3.94	4.08	4.50	4.55
(8)		HDD10.0	1416	303	243	165	82	23	3	0	0	4	68	198	327
(9)		HDD18.3	3557	562	476	421	311	197	106	25	29	110	293	444	585
(10)		CDD10.0	1053	0	0	3	21	87	159	297	290	159	34	4	0
(11)		CDD18.3	156	0	0	0	0	4	12	63	61	15	1	0	0
(12)		CDH23.3	3141	0	0	0	7	79	258	1241	1187	356	12	0	0
(13)		CDH26.7	1282	0	0	0	1	16	76	544	521	123	2	0	0
(14)	Wind	WSAvg	3.2	3.9	3.7	3.8	3.7	3.0	2.7	2.1	2.2	2.4	3.0	3.8	3.9
(15)	Precipitation	PrecAvg	518	67	51	52	42	42	33	17	21	23	39	70	69
(16)		PrecMax	712	158	135	120	118	85	88	44	97	72	84	144	158
(17)		PrecMin	378	10	2	11	5	10	5	0	0	0	13	16	16
(18)		PrecStd	87	33	30	23	23	19	18	13	20	17	24	30	34
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	10.6	12.8	18.4	24.1	28.5	32.4	35.9	36.2	33.0	25.1	16.6	10.4	
(20)		0.4% MCWB	7.7	8.2	10.0	13.5	15.7	18.7	18.2	17.5	16.3	14.3	11.2	7.4	
(21)		2% DB	8.3	10.1	14.8	20.3	25.7	28.7	33.6	33.8	30.1	21.2	13.3	8.0	
(22)		2% MCWB	5.7	6.8	8.5	11.5	14.9	17.7	17.7	16.8	15.7	12.0	9.4	5.7	
(23)		5% DB	7.0	8.5	12.5	17.5	22.8	26.2	32.1	32.0	27.5	18.2	11.3	6.4	
(24)		5% MCWB	4.8	5.7	7.6	10.2	13.8	16.1	17.3	16.4	14.6	10.8	8.2	4.5	
(25)		10% DB	5.5	7.0	10.7	14.8	20.4	23.9	29.9	29.7	24.8	15.8	9.4	4.9	
(26)		10% MCWB	3.7	4.5	6.8	8.8	12.3	15.0	16.5	15.7	13.7	9.7	6.8	3.2	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	8.2	9.0	10.8	14.6	17.1	20.7	21.7	20.3	19.7	15.3	12.4	7.9	
(28)		0.4% MCDB	10.0	11.3	15.8	22.1	26.0	28.7	31.8	32.1	28.3	23.4	15.6	9.9	
(29)		2% WB	6.1	7.3	9.5	12.2	15.6	18.5	19.7	18.0	16.8	12.9	9.9	6.1	
(30)		2% MCDB	8.0	9.5	13.8	18.8	24.0	27.3	29.5	30.6	27.4	19.1	12.8	7.8	
(31)		5% WB	4.9	6.0	8.2	10.6	14.4	16.9	18.5	17.0	15.4	11.6	8.5	4.7	
(32)		5% MCDB	6.8	8.2	11.9	16.5	21.6	24.9	28.9	29.4	25.7	17.0	10.8	6.5	
(33)		10% WB	3.9	4.7	7.0	9.3	13.1	15.5	17.4	16.2	14.1	10.2	7.1	3.5	
(34)		10% MCDB	5.5	6.7	10.3	14.3	19.1	22.8	27.6	28.0	23.4	15.0	9.2	4.9	
(35)	Mean Daily Temperature Range	MDBR	5.3	6.9	8.8	10.3	11.9	13.1	17.5	17.5	15.4	11.2	6.7	5.4	
(36)		5% DB MCDBR	6.3	8.6	11.9	14.8	16.4	18.0	21.0	21.3	20.1	15.4	8.6	6.3	
(37)		5% WB MCWBR	4.6	5.9	7.4	8.0	8.1	9.0	8.6	8.6	9.5	8.5	5.6	4.9	
(38)		5% WB MCDBR	6.0	7.9	10.6	13.5	14.4	16.8	18.4	18.7	18.2	13.4	7.9	6.1	
(39)	Clear-Sky Solar Irradiance	taub	0.273	0.276	0.290	0.309	0.329	0.327	0.331	0.338	0.318	0.303	0.277	0.271	
(40)		taud	2.520	2.560	2.564	2.497	2.454	2.482	2.468	2.438	2.474	2.529	2.560	2.507	
(41)		Ebn at Noon	843	917	947	951	936	936	927	907	899	865	832	801	
(42)		Edh at Noon	62	73	85	99	108	106	106	104	92	75	59	56	
(43)	All-Sky Solar Radiation	RadAvg	1.18	2.09	3.19	4.70	5.93	6.61	7.49	6.18	4.48	2.73	1.39	0.93	
(44)		RadStd	0.11	0.24	0.33	0.42	0.47	0.50	0.35	0.40	0.37	0.26	0.12	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	-2.20	N/A	N/A	N/A	N/A	+121	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page